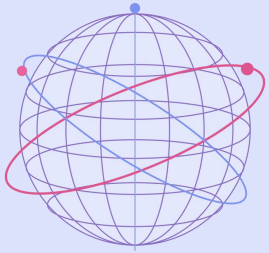
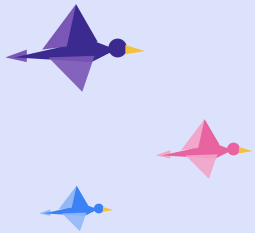
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details





Participant Details

PROJECT TITLE
Interactive QAOA Lab: Cloud-Based Quantum-Classical Comparison for Network Optimisation

TRACK
Quantum Cloud-Based Applications

TEAM NAME
Dark Scorpions

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INSTITUTION / ORGANISATION
Visvesvaraya Technological University

COUNTRY
India

Problem Statement

Students, researchers, and industry practitioners lack an accessible, interactive, cloud-based tool to:

- Test small network optimisation problems (MaxCut) on quantum hardware simulators
- Compare quantum (QAOA) vs classical (brute force) solutions side-by-side with visual evidence
- Understand the trade-offs in solution quality, circuit depth, and execution time

Current Qiskit tutorials are notebook-based, static, and disconnected from real-world context. There is no beginner-friendly web app that lets a user pick a real industry scenario, solve it quantumly, and immediately see how it compares to the classical answer.

Objectives & Novelty

Objectives:

Make quantum optimisation accessible to non-quantum students via an interactive web lab.
Provide a side-by-side quantum vs classical comparison with visual evidence
Map an abstract algorithm (QAOA) to 5 named real-world problems.
Demonstrate that quantum approaches are honest (show where they fail, not just where they win)

Novelty:

Scenario picker — user chooses VLSI / Social / Data Center / Logistics / Finance, not just "a random graph"
3D animated landing page (Three.js + GSAP) as a team portfolio, not just a notebook
Modular src/ architecture — the solver is importable as a library, not locked in a notebook
One-line hardware swap — StatevectorSampler → Sampler(mode=backend) moves from simulator to real IBM QPU without code restructuring

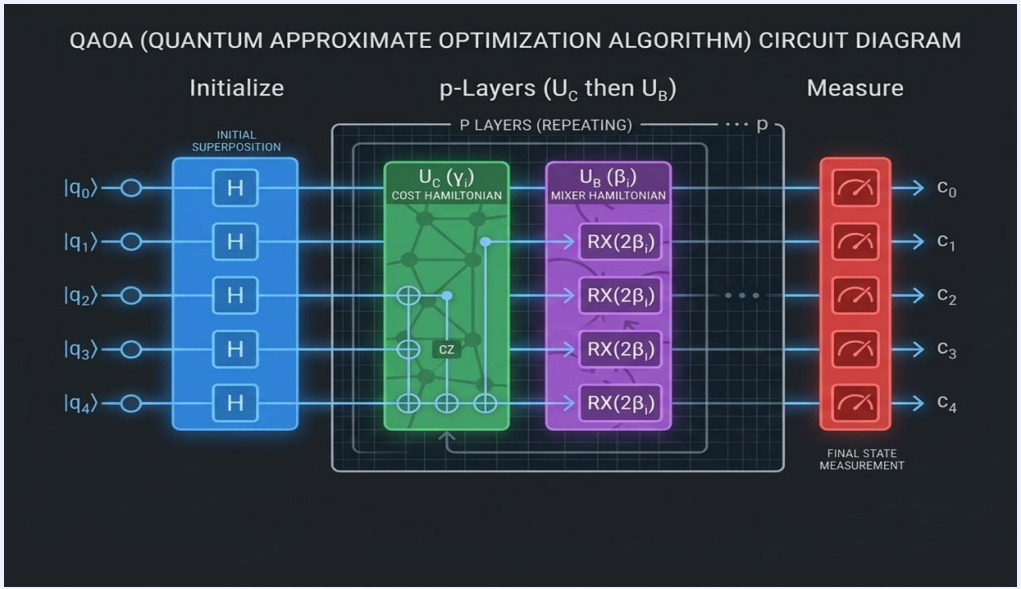
Solution & Quantum Advantage

The solution: A Streamlit web app that:

- Generates a graph from a chosen real-world scenario
- Solves MaxCut using classical brute force (2^n possibilities)
- Solves MaxCut using QAOA (variational quantum algorithm) on Qiskit Aer
- Presents both answers side-by-side with timing, accuracy, and a verdict

Aspect	Classical	Quantum
Scaling	2^n	Polynomial in circuit depth
n=4 (demo)	Instant	Instant
n=30	~1B evaluations	Still runs in seconds(on simulator
n=100	Almost Impossible	Feasible with NISQ-era Hardware.

Algorithm & Circuit Design



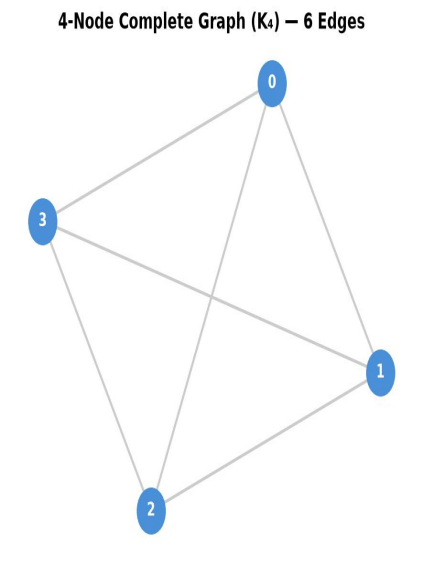
Parameters: $\gamma_1 \ldots \gamma_p$ (cost angles), $\beta_1 \ldots \beta_p$ (mixer angles) — optimized classically by COBYLA to minimize the expected energy (maximize cut value).

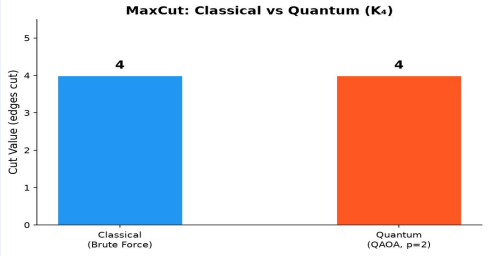
Classical baseline: Brute force — iterate all 2^n bitstrings, count cut edges, return the maximum. Exact but exponential.

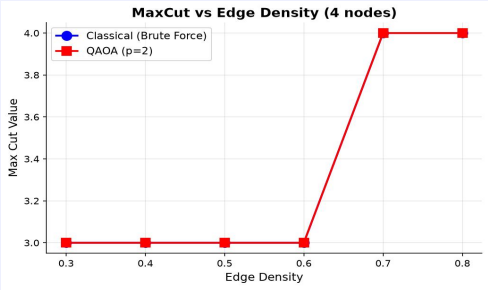
Implementation & Tech Stack

Quantum	Qiskit 2.x · QAOA (p=1,2,3) · COBYLA · StatevectorSampler
Classical	Brute-force 2^n enumeration
Graph	NetworkX 3.4
UI	Streamlit + Three.js + GSAP
Lang	Python 3.12 · venv

Results & Analysis







Metric	Classical	Quan -tum (QAO A)
Max Cut Value	4	4
Time	0.02 ms	120 ms
Method	Brute force (16 checks)	QAOA p=2 (COBYLA, 200 iter)

Impact & Real-World Applications

Industry	Problem	MaxCut Mapping
Semiconductor	VLSI floorplanning — split a chip into 2 sub-chips	Minimise inter-chip wires (cost + signal delay)
Cloud Computing	Data-center rack partitioning	Minimise cross-rack network traffic (latency)
Social Media	Community detection in user graphs	Find natural 2-group split for recommendations
Logistics	Fleet routing — split deliveries between trucks	Maximise route efficiency
Finance	Portfolio allocation — buy vs sell split	Maximise expected return

Challenges & Future Scope

Challenge	Fix
Qiskit 2.x API breaks (QuantumInstance, to_ising())	Migrated to MinimumEigenOptimizer
NetworkX 3.4 removed gnp_graph	Replaced with erdos_renyi_graph
QAOA misses optimal at n=6	Stated honestly; increased p as mitigation
Streamlit + Three.js iframe sizing	Fixed with html(height=1400)

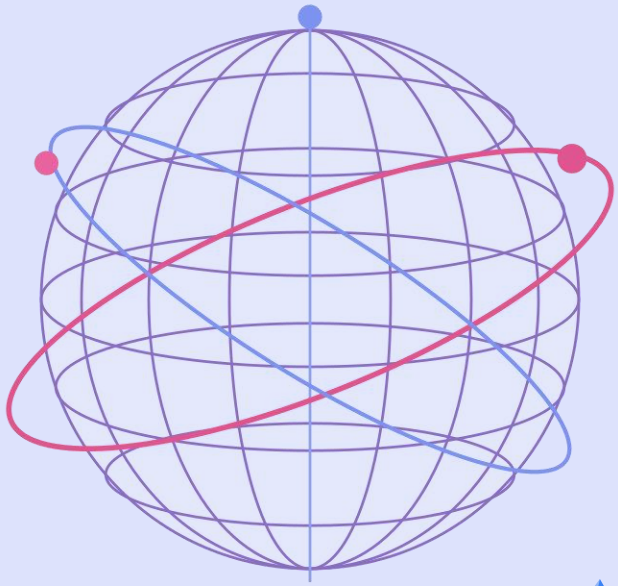
Future Scope	Effort
Run on real IBM QPU (one-line swap)	10 min
Add noise model (depolarising channel)	30 min
Scale to n=8–12 with SPSA optimiser	1 hr
Deploy live URL (Streamlit Cloud)	15 min

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.



GITHUB REPOSITORY

https://github.com/vtu-qiskit-fall-fest-2026/vtu-hc-dark-scorpions

DEMO VIDEO LINK

https://youtu.be/fnulXvdBycc?si=cnmr3JW3-_TKSpm4